

General Notions of Stochastic Processes

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Stochastic Process as a Random Function I

There are at least two views towards stochastic processes:

- ▶ A collection of (usually dependent) random variables;
- ▶ A random mapping taking values in a function space.

Let $(\Omega_t, \mathcal{A}_t)$, $t \in T$ be a collection of measurable spaces. On the Cartesian product set $\times_{t \in T} \Omega_t = \{(\omega_t; t \in T); \omega_t \in \Omega_t, \forall t \in T\}$, define the **product σ -field**:

$$\bigotimes_{t \in T} \mathcal{A}_t = \sigma\{A_t \times \times_{s \neq t} \Omega_s; A_t \in \mathcal{A}_t, t \in T\},$$

which is the σ -field generated by all the one-dimensional measurable cylinder sets.

When $(\Omega_t, \mathcal{A}_t) \equiv (\Omega, \mathcal{A})$, then the notation for the product measurable space simplifies to $(\Omega^T, \mathcal{A}^T)$, and further simplifies to $(\Omega^\infty, \mathcal{A}^\infty)$ if $T = \mathbb{N} = \{1, 2, \dots\}$, or to $(\Omega^n, \mathcal{A}^n)$ if $T = \{1, 2, \dots, n\}$.

Stochastic Process as a Random Function II

A **stochastic process** (or **random process**, or **random function**) is a measurable mapping from the measurable space $(\Omega, \mathcal{F})$ to the product space $(S^T, \mathcal{S}^T)$. The set S is called the **state space** of the process, and T is called the index set or the **time parameter space**.

The stochastic processes we will encounter in this course usually has $\mathbb{R}$ as its state space. The index set is often $T = \mathbb{Z}_+ = \{0, 1, \dots\}$ or $T = \overline{\mathbb{R}}_+$. When the index set is countable, e.g. in the former case, the process is called a discrete time process; when the index set is $\mathbb{R}$ or an interval therein, the process is called a continuous time process.

Stochastic Process as a Collection of Random Variables I

We use π_t , $t \in T$ to denote the evaluation mappings on S^T , that is, $\pi_t : S^T \rightarrow S$ defined by $\pi_t(x) = x_t$ for all $x = (x_t; t \in T) \in S^T$.

Theorem 1

Fix a measurable set $(S, \mathcal{S})$, and an index set T . Then a mapping $X : \Omega \rightarrow S^T$ is S^T -measurable iff $X_t = \pi_t \circ X : \Omega \rightarrow S$ is $\mathcal{S}$ -measurable for all $t \in T$.

Proof. The “only if” part is easy since π_t is measurable for all $t \in T$ and compositions of measurable functions are measurable. To show the “if” part, we first note that the cylinder sets $\{x \in S^T; x_t \in B\}$, $B \in \mathcal{S}$, $t \in T$ have measurable pre-images under X by the measurability of X_t . Now note that S^T is generated by the cylinder sets, and so

$$\begin{aligned} X^{-1}S^T &= X^{-1}\sigma\{\{x \in S^T; x_t \in B\}; t \in T, B \in \mathcal{S}\} \\ &= \sigma\{X^{-1}\{x \in S^T; x_t \in B\}; t \in T, B \in \mathcal{S}\} \subset \mathcal{F}. \quad \square \end{aligned}$$

Stochastic Process as a Collection of Random Variables II

A mapping X with the properties in Theorem 1 is called an S -valued process on T with paths in S^T . By the theorem it is equivalent to regard X as a collection of random elements X_t in the state space S (this is why many sources define stochastic processes as collections of random variables). Note that we use the term **random elements** to refer to measurable mappings.

Finite-Dimensional Distributions I

For any random elements ξ and η in a common measurable space (but on potentially different probability spaces), if the distributions of ξ and η are the same, then we write $\xi \stackrel{d}{=} \eta$. If X is a stochastic process on some index set T , then by the associated finite-dimensional distributions we mean the distributions of the finite-dimensional projections of X as random vectors $(X_{t_1}, \dots, X_{t_n})$, $t_1, \dots, t_n \in T$, $n \in \mathbb{N}$. The following result shows that the distribution (or **law**) of a process is completely determined by the set of finite-dimensional distributions.

Theorem 2

Fix any S, T and let X and Y be two processes on T with paths in S^T . Then $X \stackrel{d}{=} Y$ iff

$$(X_{t_1}, \dots, X_{t_n}) \stackrel{d}{=} (Y_{t_1}, \dots, Y_{t_n}), \forall t_1, \dots, t_n \in T, n \in \mathbb{N}.$$

Finite-Dimensional Distributions II

Proof. The only if part is trivial. We show the if part now. Let

$$\mathcal{D} = \{A \in \mathcal{S}^T; P(X \in A) = P(Y \in A)\}$$

and $\mathcal{C} = \{A = \{x \in \mathcal{S}^T; (x_{t_1}, \dots, x_{t_n}) \in B\}; t_1, \dots, t_n \in T, B \in \mathcal{S}^n, n \in \mathbb{N}\}$. Then, $\mathcal{C}$ is a π -system and $\mathcal{D}$ a λ -system, and $\mathcal{C} \subset \mathcal{D}$ by assumption. Hence, $\mathcal{S}^T = \sigma(\mathcal{C}) \subset \mathcal{D}$ by the monotone class theorem, which means $X \stackrel{d}{=} Y$. □

Based on the above results, when we want to specify the distribution of a stochastic process, we only need to specify its finite-dimensional distributions.

Existence of Processes I

We now state without a proof the following fundamental result of stochastic process theory, which says that for any family of “consistently” specified probability measures on finite dimensional product spaces there exists a stochastic process on the infinite-dimensional product space which has this family of probability measures as its finite-dimensional distributions. The formal concept of “consistent” here is projective. Let T be an arbitrary index set and $(S_t, \mathcal{S}_t)$, $t \in T$ be a collection of measurable spaces. Define $S_I = \times_{t \in I} S_t$ and $\mathcal{S}_I = \bigotimes_{t \in I} \mathcal{S}_t$. For a process $X = (X_t)$ on T , we also write X_I for the restriction of X to $I \subset T$. Let $\hat{T}$ and $\bar{T}$ denote the collection of finite and countable subsets of T respectively. A family of probability measures μ_I , $I \in \hat{T}$ or $\bar{T}$, is said to be **projective** if

$$\mu_J(\cdot \times S_{J \setminus I}) = \mu_I, \quad I \subset J \text{ in } \hat{T} \text{ or } \bar{T}.$$

Existence of Processes II

Theorem 3 (Kolmogorov consistency theorem)

For any collection of Borel measurable spaces $(S_t, \mathcal{S}_t)$, $t \in T$, consider a projective family of probability measures μ_I on $(S_I, \mathcal{S}_I)$, $I \in \hat{T}$. Then there exist a random element X in $S_T (= S^T)$, such that $P \circ X_I^{-1} = \mu_I$ for all $I \in \hat{T}$.

Proof. The proof is omitted (not because it is easy but because it is too technical and time-consuming). See pp. 113-116 of Kallenberg if you are interested in the proof. □

Adaptivity and Optionality I

Fix an index set $T \subset \overline{\mathbb{R}}$ and a probability space $(\Omega, \mathcal{F}, P)$. A **filtration** on T is defined as an increasing family of σ -fields $\mathcal{F}_t \subset \mathcal{F}$.

A process X is said to be **adapted** to this filtration if X_t is $\mathcal{F}_t$ -measurable for each $t \in T$. The smallest filtration with this property, namely $\mathcal{F}_t = \sigma\{X_s; s \leq t\}$, $t \in T$, is called the **induced** or **generated** filtration. It is also known as the **internal history** of the process X .

A filtration $\{\mathcal{F}_t; t \in T\}$ is said to satisfy the **usual conditions** if it is right continuous and complete, that is, $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$ for all $t \in T$, and $\mathcal{F}_0$ contains all P -null sets in $\mathcal{F}$. Note by a P -null set we mean any subset of a measurable set with probability 0 under P .

Adaptivity and Optionality II

The process X is said to be *right (left) continuous* if $t \rightarrow X_t(\omega) \equiv X(t, \omega)$ is right (left) continuous on T for each $\omega \in \Omega$ a.s. and *càdlàg* (French: “continue à droite, limite à gauche”) if it is right continuous and has limits from the left at every $t > 0$ a.s.

Two processes $\{X_t : t \in T\}$ and $\{Y_t : t \in T\}$ on the same probability space are *versions* of each other if for each t , $\mathbb{P}(X_t = Y_t) = 1$. They are said to be *indistinguishable* if $\mathbb{P}(X_t = Y_t, \forall t \in T) = 1$. Note that if X and Y are versions of each other and T is countable, then they are indistinguishable (but this is not true in general).

Adaptivity and Optionality III

A measurable mapping $\tau : \Omega \rightarrow T \cup \{\sup T\}$ is called a *stopping time* or **optional time** with respect to $\{\mathcal{F}_t\}$ if for every $t \in T$, $\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$. If T is countable, then it is clearly equivalent that $\{\tau = t\} \in \mathcal{F}_t$ for all t . The σ -algebra of events up to time τ , $\mathcal{F}_\tau$, is defined by

$$\mathcal{F}_\tau = \{A \in \mathcal{F}; A \cap \{\tau \leq t\} \in \mathcal{F}_t, t \in T\}$$

Proposition 4

Let τ_1, τ_2 be stopping times. Then $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$ are stopping times.

Proof. Since $\{\tau_1 \vee \tau_2 \leq t\} = \{\tau_1 \leq t\} \cap \{\tau_2 \leq t\}$, $\tau_1 \vee \tau_2$ is a stopping time. $\{\tau_1 \wedge \tau_2 \leq t\} = \{\tau_1 \leq t\} \cup \{\tau_2 \leq t\}$, $\tau_1 \wedge \tau_2$ is a stopping time. □

Adaptivity and Optionality IV

Lemma 5

Assume that $\{\mathcal{F}_t\}$ is a usual filtration. Then τ is a stopping time if and only if $\{\tau < t\} \in \mathcal{F}_t$ for every t .

Proof. If τ is a stopping time the $\{\tau < t\} = \cup\{\tau \leq t - 1/n\}$ and $\{\tau \leq t - 1/n\} \in \mathcal{F}_{t-1/n} \subset \mathcal{F}_t$. Conversely if $\{\tau < t\} \in \mathcal{F}_t$, $\forall t$, then $\{\tau \leq t\} = \cap_n \{\tau < t + 1/n\} \in \cap_n \mathcal{F}_{t+1/n} = \mathcal{F}_{t+} = \mathcal{F}_t$. $\square$

Proposition 6

(a) *Let τ_n be an increasing sequence of stopping times. Then $\tau = \lim_{n \rightarrow \infty} \tau_n$ is also a stopping time.*

(b) *Let $\{\mathcal{F}_t\}$ be a usual filtration and τ_n be a decreasing sequence of stopping times. Then $\tau = \lim_{n \rightarrow \infty} \tau_n$ is also a stopping time.*

Proof. If $\tau_n \uparrow \tau$, then $\{\tau \leq t\} = \cap \{\tau_n \leq t\} \in \mathcal{F}_t$. If $\tau_n \downarrow \tau$, then $\{\tau < t\} = \cup \{\tau_n < t\} \in \mathcal{F}_t$. $\square$

Adaptivity and Optionality V

Proposition 7

- (a) Let $\tau_j, j = 1, 2$, be stopping times. If $\tau_1 \leq \tau_2$, then $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$.
(b) Let $\{\mathcal{F}_t\}_{t \in T}$ satisfy the usual condition and $\{\tau_n\}$ a sequence of stopping times. If $\tau_n \downarrow \tau$, then $\cap_n \mathcal{F}_{\tau_n} = \mathcal{F}_\tau$.

Proof. To show (a), note that $A \in \mathcal{F}_{\tau_1}$ implies that $A \in \mathcal{F}$ and $A \cap \{\tau_1 \leq t\} \in \mathcal{F}_t$ for all t . By $\tau_1 \leq \tau_2$, we have

$\{\tau_2 \leq t\} \subset \{\tau_1 \leq t\}$ and thus

$A \cap \{\tau_2 \leq t\} = (A \cap \{\tau_1 \leq t\}) \cap \{\tau_2 \leq t\} \in \mathcal{F}_t$. Therefore $A \in \mathcal{F}_{\tau_2}$.

To show (b), first note $\cap_n \mathcal{F}_{\tau_n} \supset \mathcal{F}_\tau$ by (a). Let $A \in \cap_n \mathcal{F}_{\tau_n}$. Then as above, $A \cap \{\tau_n < t\} \in \mathcal{F}_t$ for each t and n . Hence

$A \cap \{\tau < t\} = \cup \{A \cap \{\tau_n < t\}\} \in \mathcal{F}_t$ and

$A \cap \{\tau \leq t\} = \cap_{n \geq 1/h} (A \cap \{\tau < t + 1/n\}) \in \mathcal{F}_{t+h}$ for all $h > 0$.

Therefore, $A \cap \{\tau \leq t\} \in \cap_{h>0} \mathcal{F}_{t+h} = \mathcal{F}_{t+} = \mathcal{F}_t$. Hence $A \in \mathcal{F}_\tau$.

